

23.1 Mass Defect & Nuclear Binding Energy

Question Paper

Course	CIEA Level Physics
Section	23. Nuclear Physics
Topic	23.1 Mass Defect & Nuclear Binding Energy
Difficulty	Easy

Time allowed: 50

Score: /38

Percentage: /100

Question 1a

Define *atomic mass unit*.

[2 marks]

Question 1b

The unified atomic mass unit u is roughly equal to 1.66×10^{-27} kg which is the mass of one nucleon.

Calculate the mass, in kg, of a nucleus of uranium-238. Give your answer to 3 significant figures.

[3 marks]

Question 1c

Einstein's Theory of Relativity showed that mass could be converted into energy, and energy into mass. This is summarised by the following equation

$$\Delta E = \Delta mc^2$$

Define the terms in the equation and give the units

(i)

E

[1]

(ii)

m

[1]

(iii)

c

[1]

[3 marks]

Question 1d

Calculate the energy released, in J, if all of the mass in the nucleus of uranium-238 were converted into energy.

[2 marks]**Question 2a**

State the meaning of the terms

(i)
binding energy,

[1]

(ii)
mass defect.

[1]

[2 marks]

Question 2b

The nuclear rest mass m_{total} of oxygen-16 ($^{16}_8\text{O}$) is 15.994914 u.

The following equation describes the relationship between mass defect Δm and the mass of the constituents of a nucleus

$$\Delta m = Zm_p + Nm_n - m_{total}$$

(i)

State the meaning of the terms Zm_p and Nm_n .

[2]

(ii)

Calculate the mass defect of oxygen-16.

$$m_p = 1.007276 \text{ u}$$

$$m_n = 1.008665 \text{ u}$$

[3]

[5 marks]

Question 2c

Use the mass defect from (b) to show that the total binding energy of a nucleus of oxygen-16 is about 2×10^{-11} J.

[3 marks]

Question 2d

Determine the binding energy per nucleon, in MeV, of oxygen-16.

[4 marks]

Question 3a

Fig. 1.1 shows the binding energy per nucleon for a number of nuclei.

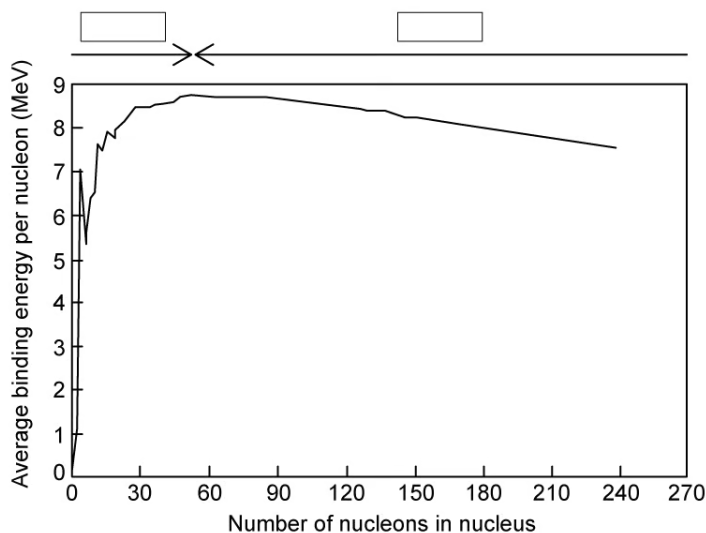


Fig. 1.1

On Fig. 1.1, mark

- the region in which fusion occurs
- the region in which fission occurs
- an X to show the location of iron-56 (${}^{56}_{26}\text{Fe}$)

[3 marks]**Question 3b**

In terms of the forces acting within the nucleus, explain why:

(i)

fusion occurs for nuclides with low nucleon numbers.

[2]

(ii)

fission occurs for nuclides with high nucleon numbers.

[2]**[4 marks]**

Question 3c

In both fission and fusion, there is a mass defect between the original nuclei and the daughter nuclei.

Complete the sentences by circling the correct word.

In fusion, the mass of the nucleus that is created is slightly **more / less** than the total mass of the original nuclei and the daughter nucleus is **more / less** stable.

In fission, an unstable nucleus is converted into more stable nuclei with a **larger / smaller** total mass. In both cases, this difference in mass, the mass defect, is equal to the binding energy that is released.

Fission / Fusion releases much more energy per kg than **fission / fusion**. The greater the increase in binding energy, the **more / less** energy is released.

[4 marks]

Question 3d

The graph in Fig. 1.2 shows the binding energy per nucleon in MeV plotted against nucleon number, A .

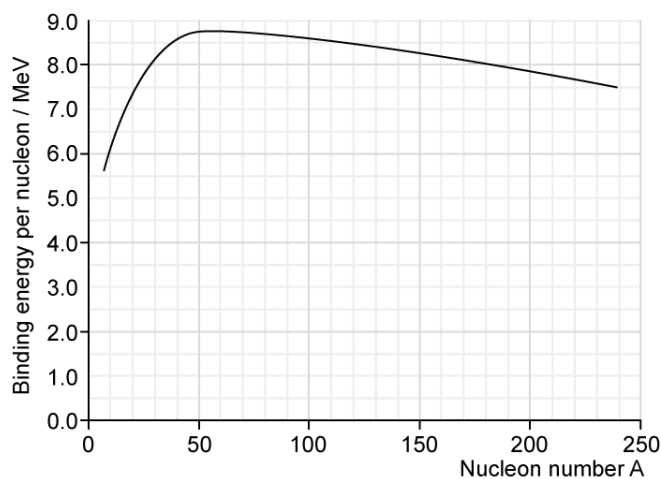


Fig. 1.2

Use Fig. 1.2 to find the binding energy of the following nuclei:

(i)
platinum-190.

[1]

(ii)
silicon-28.

[1]

(iii)
tellurium-120.

[1]

[3 marks]

Model Answers: Easy

1a

(a) The atomic mass unit is defined as:

- A twelfth / $\frac{1}{12}$ of the mass; [1 mark]
- Of a (neutral) atom of carbon-12; [1 mark]

[Total: 2 marks]

This is the precise definition of the atomic mass unit - make sure you learn it in terms of how it relates to carbon-12. It is equal to 1.66×10^{-27} kg, which is roughly equal to a proton or neutron, hence, this is what you will need to use in calculation questions.

1b

(b) Calculate the mass of a nucleus of uranium-238:

List the known quantities:

- Mass number of uranium-238 = 238
- Atomic mass unit, $u = 1.661 \times 10^{-27}$ kg

Determine the mass of uranium-238 in kg:

- Mass of uranium-238 = $238 u = (1.661 \times 10^{-27}) \times 238$ [1 mark]
- Mass of uranium-238 = 3.95318×10^{-25} kg [1 mark]
- Mass of uranium-238 = 3.95×10^{-25} kg (3 s.f.) [1 mark]

[Total: 3 marks]

Exam questions commonly ask you to "estimate" the mass of a nuclide, but don't let this throw you off, as this is simply referring to the fact that the atomic mass unit is **approximately** equal to its total mass in a.m.u.

1c

(c)

(i) E represents the:

- Energy **AND** (measured in) joules / J; [1 mark]

(ii) m represents the:

- Mass **AND** (measured in) kilograms / kg; [1 mark]

(iii) c represents the:

- Speed of light **AND** (measured in) metres per second / m s^{-1} ; [1 mark]

[Total: 3 marks]

1d

(d) Calculate the energy released by a nucleus of uranium-238

List the known quantities:

- Mass of uranium-238 nucleus, $\Delta m = 3.95 \times 10^{-25} \text{ kg}$ (from (b))
- Speed of light, $c = 3.00 \times 10^8 \text{ m s}^{-1}$

Substitute the known values into the equation:

- Mass-energy equivalence: $\Delta E = \Delta mc^2$
- $\Delta E = (3.95 \times 10^{-25}) \times (3.00 \times 10^8)^2$ [1 mark]
- $\Delta E = 3.56 \times 10^{-8} \text{ J}$ (3 s.f.) [1 mark]

[Total: 2 marks]

2a

(a)

(i) Binding energy is defined as:

- The energy needed / required to separate the nucleons of a nucleus **OR** the energy released when a nucleus is formed from its nucleons; [1 mark]

(ii) Mass defect is defined as:

- The difference between the mass of a nucleus, and the sum of the masses of the nucleons; [1 mark]

[Total: 2 marks]

Remember that a nucleus's binding energy is required to **combine** the individual protons and neutrons into the nucleus. As a result, some of the mass of the protons and neutrons is sacrificed and converted into energy (using $E = mc^2$). This loss of mass is the mass defect.

However, when defining binding energy and mass defect in the exam, use the keywords and stick to the definitions outlined above to ensure you get the marks. It is ok to say 'neutrons and protons' instead of nucleons, but you need to be specific: for example, saying 'constituent parts' isn't enough to get you the marks.

2b

(b)

(i) The meaning of the terms is:

- Z = proton number / number of protons **AND** m_p = proton rest mass
OR
- Zm_p = total mass of the protons in the nucleus; [1 mark]
- N = number of neutrons **AND** m_n = neutron rest mass
OR
- Nm_n = total mass of the neutrons in the nucleus; [1 mark]

(ii) Calculate the mass defect of oxygen-16:

List the known quantities:

- Nuclear rest mass of oxygen-16 ($^{16}_8\text{O}$) = 15.994914 u
- Proton rest mass, m_p = 1.007276 u
- Neutron rest mass, m_n = 1.008665 u

Determine the numbers of protons and neutrons in oxygen-16

- From $^{16}_8\text{O}$:
- Number of protons, $Z = 8$
- AND**
- Number of protons, $N = 16 - 8 = 8$ [1 mark]

Substitute the known values into the equation:

- $\Delta m = Zm_p + Nm_n - m_{total}$
- $\Delta m = (8 \times 1.007276 \text{ u}) + (8 \times 1.008665 \text{ u}) - 15.994914 \text{ u}$ [1 mark]
- $\Delta m = 0.132614 \text{ u}$ [1 mark]

[Total: 5 marks]

The equation $\Delta m = Zm_p + Nm_n - m_{total}$ is **not** in the data booklet, but you do need to know how to calculate Δm .

You may prefer to memorise the equation, but the best approach is to remember that the mass defect is the **sum of the individual rest masses**, minus the **total rest mass of the nucleus** in question.

2c

(c) Show that the total binding energy of a nucleus of oxygen-16 is about $2 \times 10^{-11} \text{ J}$:

List the known quantities:

- Mass defect of oxygen-16, $\Delta m = 0.132614 \text{ u}$ (from **(b)**)
- Speed of light, $c = 3.00 \times 10^8 \text{ m s}^{-1}$
- Atomic mass unit, $u = 1.66 \times 10^{-27} \text{ kg}$

Substitute the known values into the mass-energy equivalence equation to calculate ΔE :

- Mass-energy equivalence $\Delta E = \Delta mc^2$
- Mass defect (in J): $\Delta m = (0.132614) \times (1.66 \times 10^{-27})$ [1 mark]
- Binding energy: $\Delta E = (0.132614) \times (1.66 \times 10^{-27}) \times (3.00 \times 10^8)^2$ [1 mark]

- Binding energy: $\Delta E = 1.981 \times 10^{-11} \text{ J}$ [1 mark]

[Total: 3 marks]

In this topic, you are working with **lots** of decimal places and lots of scientific notation. This has the potential for calculation errors to creep in, especially under the

- Binding energy: $\Delta E = 1.981 \times 10^{-11} \text{ J}$ [1 mark]



[Total: 3 marks]

In this topic, you are working with **lots** of decimal places and lots of scientific notation. This has the potential for calculation errors to creep in, especially under the pressure of the exam.

It is best practice to double-check **every** number you input into your calculator so that you can catch any such errors as you are working. Due diligence like this will save you time and ensures you don't make any silly mistakes.

2d

(d) The binding energy per nucleon, in MeV, of oxygen-16 is:

List the known quantities:

- Binding energy of oxygen-16, $\Delta E = 1.981 \times 10^{-11} \text{ J}$ (from **(c)**)
- Number of nucleons in oxygen-16 = 16
- $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$

Determine the binding energy per nucleon:

- Binding energy per nucleon = $\frac{\text{Binding energy}}{\text{Number of nucleons}}$
- Binding energy per nucleon = $\frac{(1.981 \times 10^{-11})}{16}$ [1 mark]
- Binding energy per nucleon = $1.238 \times 10^{-12} \text{ J}$ [1 mark]

Convert the energy into MeV:

- Binding energy per nucleon = $\frac{1.238 \times 10^{-12}}{1.60 \times 10^{-19}}$ [1 mark]
- Binding energy per nucleon = $7\,737\,500 = 7.74 \text{ MeV}$ [1 mark]

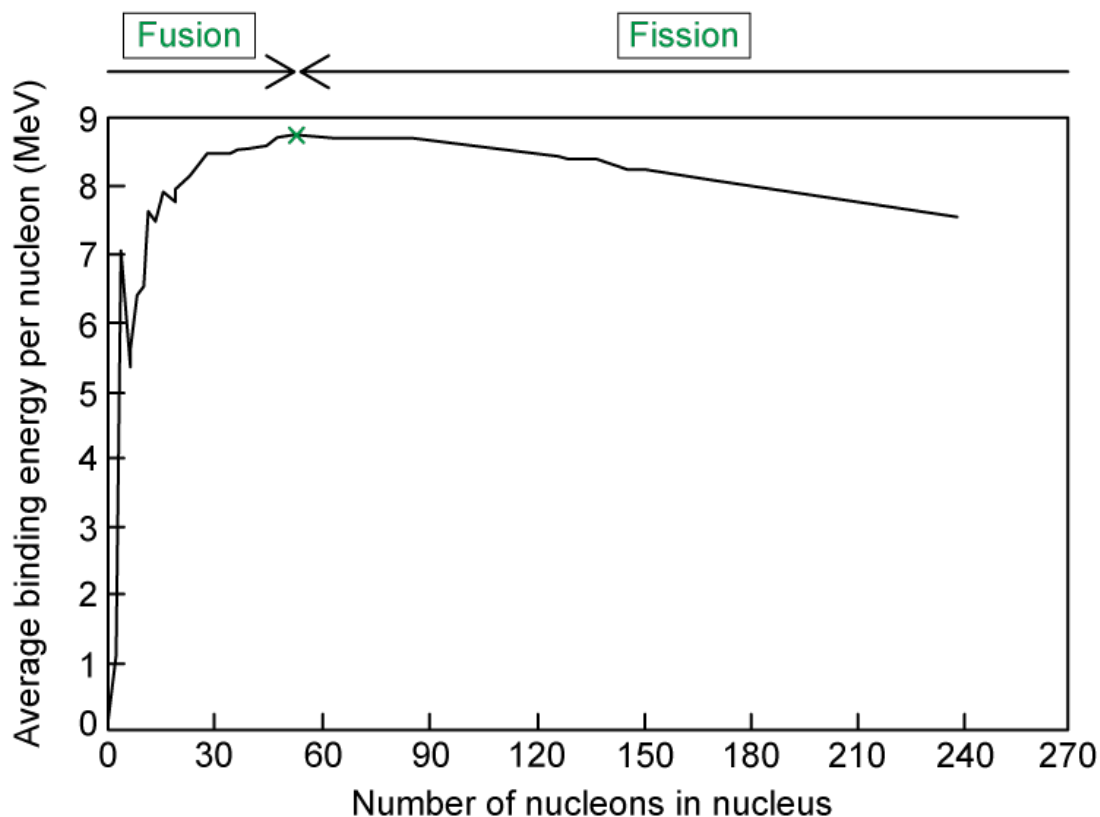
[Total: 4 marks]

The binding energy per nucleon is the **total** binding energy **divided** by the nucleon number. In this case, since the nucleon number is 16, the binding energy per nucleon is found by **dividing** the total binding energy by 16.

3a

(a) The correct labels are as follows:

One mark for each correct label:



- Fusion labelled on the left hand side of the graph; [1 mark]
- Fission labelled on the right hand side of the graph; [1 mark]
- Iron located with an X on the highest dot; [1 mark]

[Total: 3 marks]

The binding energy per nucleon is the total binding energy of a nucleus divided by the number of nucleons. This is an important quantity because it tells us about the **stability** of the nucleus.

The greater the binding energy per nucleon, the greater the stability of the nucleus because more energy is required to remove a nucleon from the nucleus.

The most important feature that you need to remember about the graph is that iron has the **highest binding energy per nucleon**, which makes it the most **stable** element.

3b

(b)

(i) Fusion occurs for nuclides with low nucleon numbers because:

- Attractive strong nuclear forces between nucleons dominate; [1 mark]
- Over repulsive electrostatic forces between protons; [1 mark]

(ii) Fission occurs for nuclides with high nucleon numbers because:

- Repulsive electrostatic forces between protons (begin to) dominate; [1 mark]
- These forces (tend to) break apart the nucleus; [1 mark]

[Total: 4 marks]

Nuclei with low nucleon numbers tend to have a lower binding energy per nucleon, hence, they are generally less stable.

This means the lightest elements have weaker electrostatic forces and are the most likely to undergo **fusion**.

For nuclei with high nucleon numbers, the general binding energy per nucleon is high and gradually decreases with increasing nucleon number.

This means the heaviest elements are the most unstable and likely to undergo **fission**.

3c

(c) The correct sentences are as follows:

One mark for each correct sentence:

- In fusion, the mass of the nucleus that is created is slightly **less** than the total mass of the original nuclei and the daughter nucleus is **more** stable; [1 mark]
- In fission, an unstable nucleus is converted into more stable nuclei with a **smaller**

total mass; [1 mark]

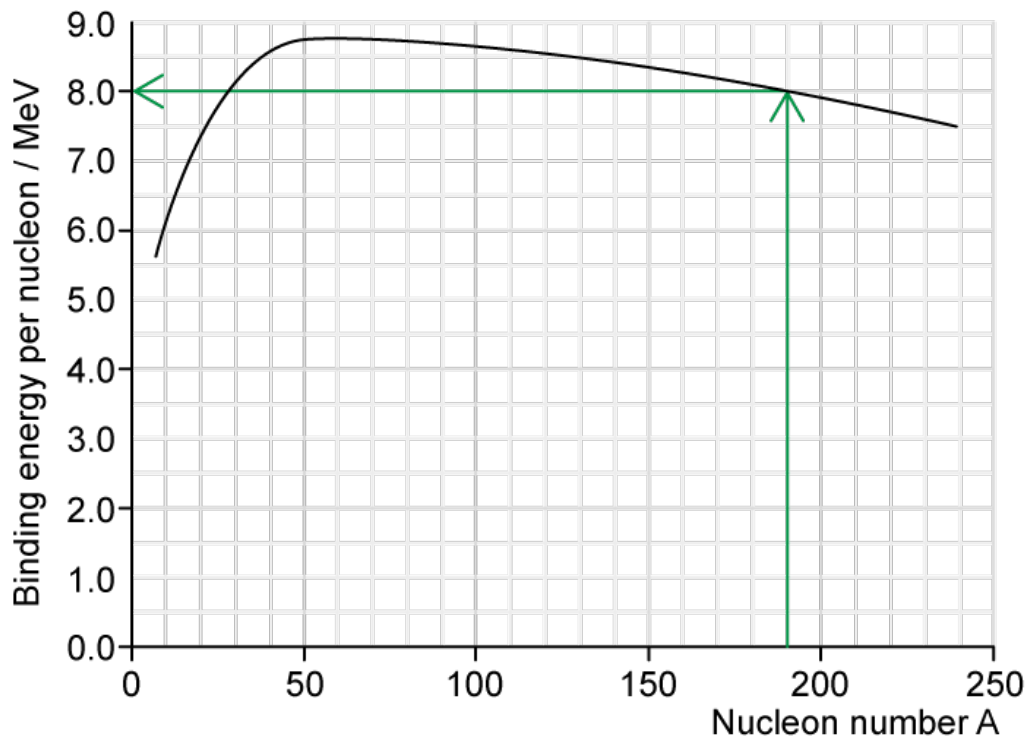
- In both cases, this difference in mass, the mass defect, is equal to the binding energy that is released.
- **Fusion** releases much more energy per kg than **fission**; [1 mark]
- The greater the increase in binding energy, the **more** energy is released; [1 mark]

[Total: 4 marks]

3d

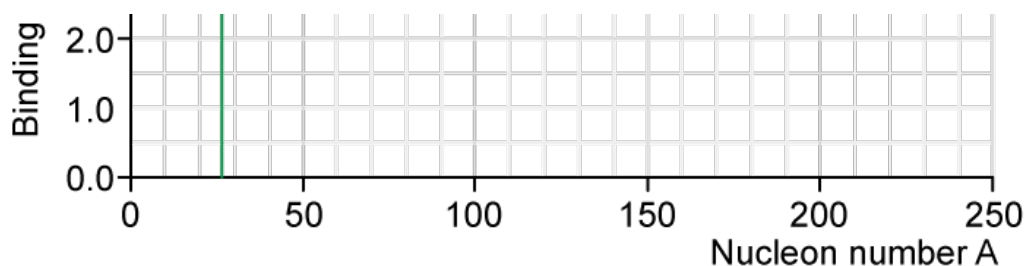
(d) The graph can be used to find the binding energy by drawing arrows as follows:

(i) Platinum–190:



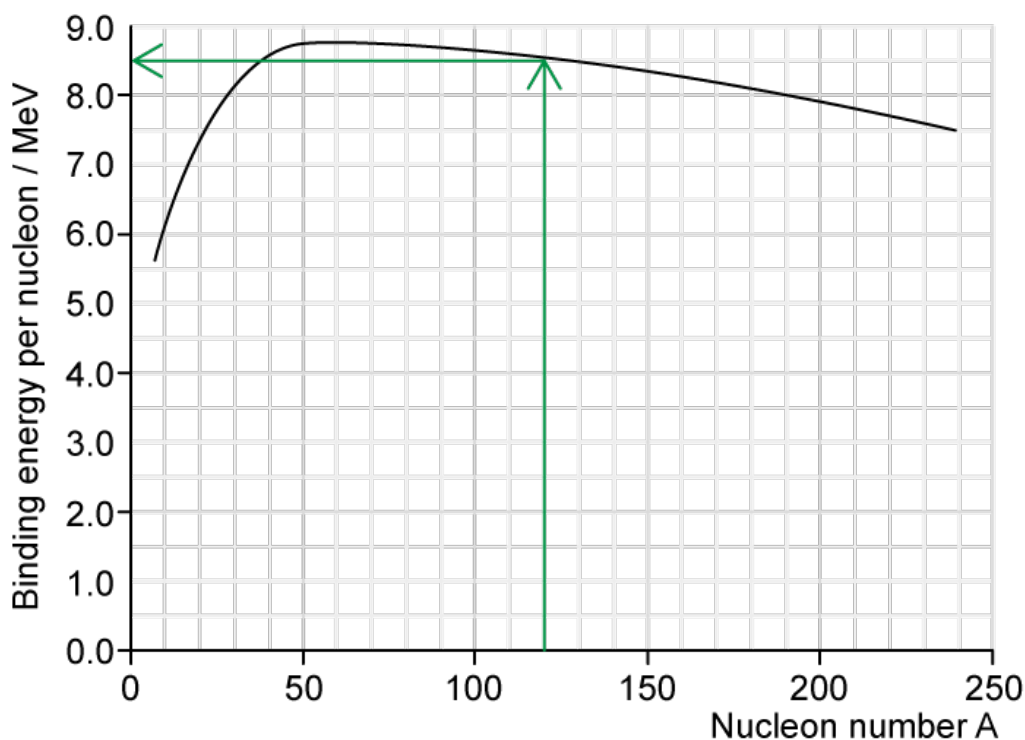
- (From graph) the binding energy for Platinum–190 is 8.0 MeV [1 mark]

(ii) Silicon–28:



- (From graph) the binding energy for Silicon-28 is 8.0 MeV [1 mark]

(iii) Tellurium-120:



- (From graph) the binding energy for Tellurium-120 is 8.5 MeV [1 mark]

[Total: 3 marks]

It is fairly straightforward to read the binding energy per nucleon from this graph for any given nucleon number as long as you understand what the axes are showing.

However, you do need to be familiar with some of the features of this graph:

- Iron (Fe) has the **greatest** binding energy per nucleon making it the most **stable** element
- Hydrogen, H, has a binding energy of zero because it only has one nucleon (it does not need to bind to anything else)
- The curve rises sharply for nuclides with low nucleon numbers, A

- The curve drops gently for nuclides with high nucleon numbers, A
- Helium (He), Carbon (C), and Oxygen (O), do not fit the general trend of the graph

